MANAGING PORTFOLIO RISK

We have previously discussed adaptation techniques to manage risk from a portfolio optimization perspective following our AIM and PIM methodologies using real-time re-optimization. In this section we discuss three techniques to determine how to evaluate potential deviation tactics for individual stocks.

These are:

1. Minimize Trading Risk
2. Maximize Trading Opportunity
3. Program-Block Decomposition

From the investor's perspective, these techniques provide: improved algorithmic trading rules, better specification of market and limit orders, and appropriate utilization of non-traditional trading venues such as crossing networks and dark pools where liquidity is not transparent. These criteria have been stated in *Optimal Trading Strategies* (2003) and in *Algorithmic Trading Strategies* (2006). We expand on those findings and apply them to today's portfolio trading algorithm needs.

Residual Risk Curve

The total dollar risk in a trade period for a portfolio is:

$$Risk_{\$}(t) = \sqrt{r_t^t C r_t} \tag{9.25}$$

where r_t is the residual share vector at time t and C is the side adjusted covariance matrix scaled for the length of the trading interval.

The residual risk curve shows how the total portfolio risk will change as we change the number of shares of a particular stock and hold the amount of all other shares constant.

From the first and second derivatives of Equation 9.25 we find that the residual risk curve is a convex function with a single minimum value at:

$$r_{i,\min} = \frac{-1}{\sigma_i^2}\sum_{j \neq i} r_j \sigma_{ij} \tag{9.26}$$

This minimum value could be either more or less than the current number of shares of the stock in the portfolio. If the minimum value is less than the current position traders could reduce portfolio risk by trading shares and reducing the holding size. If the minimum value is greater than the current position traders could reduce portfolio risk by adding shares to